SANGJUN SON

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INTRODUCTION

Software Engineer at Moloco. Interested in engineering (Backend & DevOps) and business development, with a keen interest in exploring emerging technologies and actively participating in discussions with others.

WORKING EXPERIENCES

Software Engineer, *Moloco, Next - Serving & Supply* May. 2025 - Now

- Architected real-time bidding optimization features for RMG platform, including state-level bid modifiers and event-aware creatives for advertisers such as DraftKings, Fanatics Sportsbook, Underdog Fantasy, and Kalshi. Improved regional budget allocation and trending-event relevance through production A/B experiments.
- Led OpenRTB 2.6 CTV integrations across six major SSPs (INDEX Exchange, Magnite, FreeWheel, AdRise, Publica, and SpringServe), enabling direct supply expansion and laying the foundation for Dynamic Ad Podding and richer ad-break aware bidding.
- Architected a zone-aware gRPC inference platform for multi-zone GKE clusters, redesigning service discovery and load balancing to eliminate connection stickiness, improve inference scalability, and significantly reduce serving infrastructure costs.

Site Reliability Engineer, *Moloco, Ads Engineering* Jul. 2022 - Apr. 2025

- Dynamic resource scaling system: Developed a scaling system for GCP and Kubernetes clusters based on event schedules. Successfully managed 1.5M RPS during the 2023/2024 IPL and cut infrastructure costs for major clients (Viacom18, JioCinema, TVING) through optimized use of cloud resources.
- Efficiency of the CI/CD pipeline and image optimization: Optimized Docker images, cutting vulnerabilities by 80% and image sizes by 5x. Achieved faster CD with 10x image pulling speed. Automated release notes and dashboards, enhancing visibility across 10K+ releases.
- A/B testing infrastructure for microservices: Built a staging environment isolated from production using Envoy-based traffic mirroring, preventing regressions from impacting live services. Supports 1K+ experiments with a monitoring dashboard for stability and rapid intervention.

Software Engineer Intern, *NFTBank, Backend Team* Dec. 2021 - Feb. 2022

- Scholar Resume for Axie Infinity: Created a single-page application (Scholar Resume) for Axie Infinity scholarship applicants, showcasing scholars' game career, including ranks, MMR, and earned history. Introduced an automated payout service, streamlining the distribution of earned SLP rewards to players.

Google Software Engineering Intern, *Google Korea LLC., Desktop Search Team* Jun. 2021 - Sep. 2021

- Automatic I2F and nesting config generation for hOSRP to diOSRP conversion: Designed an internal tool for efficient development in OSRP migration.

RESEARCH EXPERIENCES

Data Mining Laboratory, *Seoul National University* Nov. 2019 - Feb. 2021
Research Intern (Advisor: Prof. U Kang)

Real-Time Ubiquitous Systems Laboratory, *Seoul National University* Jul. 2017 - Feb. 2018
Research Intern (Advisor: Prof. Chang-Gun Lee)

EDUCATION

B.S. in Computer Science and Engineering, *Seoul National University* Mar. 2016 - Aug. 2022
Interdisciplinary Major in Entrepreneurship, *Seoul National University* Mar. 2021 - Aug. 2022

PUBLICATIONS

"**DAO-CP: Data-adaptive online CP decomposition**," *PLOS ONE 2022*,
Sangjun Son*, Yongchan Park*, Minyong Cho, and U Kang

"**Gtensor: Fast and Accurate Tensor Analysis System using GPUs**," *CIKM 2020*,
29th ACM International Conference on Information and Knowledge Management, Virtual Event, Ireland,
Dawon Ahn, **Sangjun Son**, and U Kang

SOFTWARE and PROJECTS

glnk.dev, *Custom Go-Link Shortener Inspired by Google's go/link* Fall. 2024
Built a self-serve short URL redirection system with personalized domains (e.g., `gcp.glnk.dev`) and custom paths. Supports over 100+ links across users. Implemented GitHub Pages-based redirection with automated domain provisioning and user registration.

LinkedArt, *College of Art Exhibition Archive Platform* Spring. 2022
Created an artwork sales platform connecting buyers with artists and established a networking community for art college students. Utilized Next.js and NestJS for the front-end and back-end frameworks, with PostgreSQL as the database, and implemented a CI/CD pipeline using Vercel and Heroku for deployment.

MopReM: Moiré Pattern Removal for Mobile, *Texts/Diagrams on Single-colored Background* Fall. 2021
Established a efficient module for mobile cameras specialized in demoireing recaptured screen materials.

Deep Learning-based Wrinkle Detection, *Morpheus3D* Fall. 2020
Built new models to segment wrinkle parts in 3D scanned face images by exploiting state-of-the-art methods.

ABC: Art with Block-Chain, *Media-art Platform* Spring. 2020
Designed a platform where any creator can upload their own media arts and increase profits. Implemented smart contract on the ERC-721 token that records artwork metadata and p5.js-based content on blockchain.

TEACHING EXPERIENCES

International Students Integrated Peer Tutoring Program, *Undergraduate Student Tutor* Spring. 2021
Data Structures & Algorithm Fundamentals, SNU Gwanak Residence Halls

Digital Computer Concept and Practice, *Lab Class Lecturer* Fall. 2020
Introduction to Python and Its Application, Dept. of Computer Science and Engineering

Basic Calculus 1, *Undergraduate Student Tutor* Spring. 2020
TA Office of the Department of Mathematical Sciences

HONORS and AWARDS

ACM ICPC Regional Contest Seoul, *Team Gogle*, 15th place Nov. 2021

Korea Olympiad in Informatics, *National Programming Contest for High School Students* May. 2015
Silver medal, 3rd place

EXTRA-CURRICULAR ACTIVITIES

WD Partners, *Consulting Firm providing Indoor Ventilation Solution* Jun. 2021 - Dec. 2021
Demonstrated optimal condition for high ventilation efficiency through CFD analysis for pollutants. (e.g., fine dust and droplets containing viruses).

Decipher, *Blockchain Research Group in Seoul National University* Mar. 2020 - Aug. 2020
Attended weekly seminars about various blockchain topics as a member of StuDeFi.
Designed a donation platform, AID-U for contributing student education expenses.